

Student Mental Health Monitoring System

SE LAB Experiment-02B BATCH-D

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1 LITERATURE REVIEW

1.1 Protecting Student Mental Health with a Context-Aware Machine Learning Framework for Stress Monitoring

Source: Protecting Student Mental Health with a Context-Aware Machine Learning Framework for Stress Monitoring

Link: <https://arxiv.org/abs/2508.01105>

Problem Solved: Universities struggle to detect student stress early because mental health issues are often hidden and not reported by students themselves. Traditional methods rely only on counseling visits or surveys, which are reactive instead of proactive.

Technique Used: The researchers developed a machine learning framework that collects contextual information such as academic workload, lifestyle habits, and self-reported feelings. Advanced algorithms like PCA and ensemble classifiers were used to analyze patterns and predict stress levels automatically.

Prominent Results: The system achieved extremely high accuracy (around 99%) in identifying stressed students, proving that data-driven monitoring can replace manual observation. It showed that intelligent analytics can serve as an effective early-warning mechanism.

What Can Be Improved: The model was tested mainly on experimental datasets and not fully integrated into real educational platforms. Real-time mobile data collection and larger student samples would make it more practical.

1.2 Student Mental Health Screening via Fitbit Data Collected During the COVID-19 Pandemic

Source: Student Mental Health Screening via Fitbit Data Collected During the COVID-19 Pandemic

Link: <https://arxiv.org/abs/2601.16324>

Problem Solved: Mental health assessment normally depends on self-report surveys, which are irregular and subjective. There is a need for continuous, objective monitoring of students' emotional well-being.

Technique Used: The study used wearable Fitbit devices to collect physiological data such as heart rate, sleep duration, and daily activity. Machine learning algorithms processed this data to identify signs of anxiety, depression, and stress.

Prominent Results: The models were able to predict different mental health conditions with strong performance, proving that wearable sensors can act as reliable screening tools for students.

What Can Be Improved: The system was tested on a limited group during the pandemic period only. More diverse student populations and additional sensors could further strengthen predictions.

1.3 Intelligent Classification and Prediction of Students' Mental Health in Online Learning Environments

Source: Intelligent Classification and Prediction of Students' Mental Health in Online Learning Environments

Link: <https://www.nature.com/articles/s41598-025-04681-2>

Problem Solved: With the rise of online education, large amounts of student interaction data are generated, but institutions lack tools to analyze this data for mental health assessment.

Technique Used: The authors applied natural language processing and machine learning to analyze students' online messages, assignments, and communication patterns. Psychological text features were extracted using LIWC and classified using optimized algorithms.

Prominent Results: The system achieved very high prediction accuracy in identifying anxiety, depression, and stress from online behavior, showing that digital footprints can reveal mental health status.

What Can Be Improved: The approach focuses mainly on online learning data and may not work well for offline students. Integration with mobile apps and counselor dashboards would make it more useful.

1.4 Research on Monitoring and Intervention Systems for College Students' Mental Health Based on AI

Source: Research on Monitoring and Intervention Systems for College Students' Mental Health Based on AI

Link: <https://ojs.bbwpublisher.com/index.php/JCER/article/view/9533>

Problem Solved: Student mental health data is usually scattered across different sources like counselors, academic records, and surveys, making holistic monitoring difficult.

Technique Used: An AI-based monitoring system was designed that combines questionnaire responses, behavioral data, and psychological scales. Decision tree algorithms were used to classify students into risk categories.

Prominent Results: The integrated system improved accuracy in identifying at-risk students and helped counselors take timely intervention actions.

What Can Be Improved: The system relies mainly on periodic data collection; adding real-time alerts and mobile integration would make it more effective.

1.5 Undergraduate Students' Mental Health Remote Monitoring System: End-User Usability Testing Study

Source: Undergraduate Students' Mental Health Remote Monitoring System: End-User Usability Testing Study

Link: <https://journals.iium.edu.my/kict/index.php/IJPCC/article/view/283>

Problem Solved: Even when mental health systems exist, students often do not use them because the interfaces are complex or not user-friendly.

Technique Used: A web-based remote monitoring platform was developed where students can record mood, fill questionnaires, and communicate with counselors. Usability testing was performed with real users.

Prominent Results: The study showed that a simple, well-designed system can effectively help students track their mental well-being and encourage them to seek help.

What Can Be Improved: The system focused mainly on usability; adding AI-based prediction and automated alerts would make it more intelligent.

1.6 Automated Assessment of Student Mental Health Through Image Processing Technology

Source: Automated Assessment of Student Mental Health Through Image Processing Technology

Link: <https://www.iieta.org/journals/ts/paper/10.18280/ts.410528>

Problem Solved: Most monitoring systems depend only on surveys or text input, which cannot capture real emotional expressions of students.

Technique Used: The study applied image processing and facial expression recognition to analyze micro-expressions and body language for assessing mental states.

Prominent Results: The proposed method demonstrated that emotional conditions like stress and anxiety can be detected automatically without asking students directly.

What Can Be Improved: Image-based monitoring alone is not sufficient; combining it with academic and behavioral data would give more reliable results.

2 ANNOTATED BIBLIOGRAPHY

2.1 Protecting Student Mental Health with a Context-Aware Machine Learning Framework for Stress Monitoring

Citation: Ovi, M. S., Hossain, J., Rahi, M. R. A., & Akter, F. (2025). Protecting Student Mental Health with a Context-Aware Machine Learning Framework for Stress Monitoring. arXiv.

Retell: This paper proposes a context-aware machine learning framework designed to monitor student stress levels automatically. The authors collect diverse contextual data such as academic workload, lifestyle patterns, and emotional self-reports. Advanced machine learning techniques are applied to analyze these features and identify stress indicators without requiring constant manual input from students.

Reflect: The study clearly demonstrates that mental health monitoring becomes significantly more accurate when multiple data sources are combined. It highlights the limitation of traditional survey-based systems and shows how artificial intelligence can provide early warnings before stress develops into serious mental health disorders.

Relate: This research is highly relevant to our Student Mental Health Monitoring System, as it supports the idea of automated and proactive stress detection. The context-aware approach can be directly adapted to our project to improve early identification and timely intervention.

2.2 Student Mental Health Screening via Fitbit Data Collected During the COVID-19 Pandemic

Citation: Lopez, R., Shrestha, A., Tlachac, M. L., et al. (2026). Student Mental Health Screening via Fitbit Data Collected During the COVID-19 Pandemic. arXiv.

Retell: This study explores the use of wearable Fitbit devices to screen student mental health conditions such as stress, anxiety, and depression. Physiological data including heart rate, sleep quality, and physical activity levels were continuously collected and analyzed using machine learning models to predict mental health status.

Reflect: The research emphasizes the importance of objective and continuous monitoring instead of relying solely on subjective self-reports. It demonstrates how wearable technology can detect subtle physiological changes associated with mental health issues, making screening more reliable and timely.

Relate: The findings are highly applicable to our project as wearable data integration can enhance the accuracy of mental health monitoring. This approach can be considered as a future extension to provide real-time and personalized insights for students.

2.3 Intelligent Classification and Prediction of Students' Mental Health in Online Learning Environments

Citation: Xu, X., & Zhang, T. (2025). Intelligent Classification and Prediction of Students' Mental Health in Online Learning Environments. Scientific Reports.

Retell: The paper focuses on analyzing students' online learning behavior to predict mental health conditions. It uses natural language processing techniques to examine online discussions, assignments, and communication patterns, extracting psychological features for classification.

Reflect: This research proves that digital learning behavior and textual data are strong indicators of students' emotional and psychological states. It also highlights the growing importance of analyzing online education platforms, especially with the increasing adoption of remote learning.

Relate: Our Student Mental Health Monitoring System can adopt similar NLP-based techniques to analyze academic and communication data. This will allow the system to identify mental health risks even when students are not physically present on campus.

2.4 Research on Monitoring and Intervention Systems for College Students' Mental Health Based on AI

Citation: Lyu, M. (2025). Research on Monitoring and Intervention Systems for College Students' Mental Health Based on AI. Journal of Contemporary Educational Research.

Retell: This article presents an AI-driven system that integrates multiple sources of student data, including academic records, questionnaires, and behavioral patterns, to monitor mental health and classify students into different risk levels.

Reflect: The study highlights the importance of intervention along with monitoring. It demonstrates how AI systems can support counselors by providing actionable insights and prioritizing students who need immediate attention.

Relate: This research closely aligns with our project's objective of building a comprehensive monitoring and intervention system. The risk classification approach can be used to design alert mechanisms and counselor dashboards in our system.

2.5 Undergraduate Students' Mental Health Remote Monitoring System: End-User Usability Testing Study

Citation: Liaw, M. A. W., et al. (2025). Undergraduate Students' Mental Health Remote Monitoring System: End-User Usability Testing Study. International Journal on Perceptive and Cognitive Computing.

Retell: This study evaluates a remote mental health monitoring platform designed for undergraduate students. The system allows students to track mood, complete psychological assessments, and communicate with counselors, while usability testing evaluates user experience.

Reflect: The research highlights that even the most advanced systems fail if they are not user-friendly. It emphasizes the importance of intuitive design, accessibility, and ease of interaction to encourage consistent student engagement.

Relate: Our project also aims to create a student-centric system. The usability principles discussed in this paper will guide the design of our dashboard and interaction flow to ensure high adoption rates.

2.6 Automated Assessment of Student Mental Health Through Image Processing Technology

Citation: Li, F. (2024). Automated Assessment of Student Mental Health Through Image Processing Technology. IIETA.

Retell: This paper introduces an innovative approach to assessing student mental health using image processing and facial expression analysis. The system detects emotional states by analyzing facial micro-expressions and body language.

Reflect: The study demonstrates that non-verbal cues can reveal emotional conditions that students may not openly express. It highlights the potential of passive monitoring techniques in mental health assessment.

Relate: This technique could be integrated as a future enhancement to our Student Mental Health Monitoring System. Combining image-based analysis with behavioral and academic data would significantly improve system accuracy.